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Rapid initial recovery and long-term persistence of a bee community in a former landfill

THOMAS M. ONUFERKO,[1] DIMITRI A. SKANDALIS,[2] RODRIGO LEÓN CORDERO[3] and MIRIAM H. RICHARDS[4] [1]Department of Biology, York University, Toronto, ON, Canada, [2]Department of Zoology, University of British Columbia, Vancouver, BC, Canada, [3]School of Environmental Sciences, University of Guelph, Guelph, ON, Canada and [4]Department of Biological Sciences, Brock University, St. Catharines, ON, Canada

Abstract. 1. The effects of habitat restoration are usually studied using cross-sectional comparisons of species assemblages among sites of various ages or disturbance levels. Longitudinal studies, however, are necessary for detecting long-term responses to habitat restoration and for understanding annual demographic variation.

2. To investigate the time course of bee community restoration in sites previously made uninhabitable by anthropogenic disturbance, we studied a former landfill site for 10 years from initial revegetation in 2003 until 2013, comparing two restored sites with three nearby, undisturbed control sites. We used permutational multivariate analysis of variance and generalised additive mixed models to investigate how bee abundance and species richness varied over time (years), between seasons and between restoration levels.

3. Landfill restoration and the creation of foraging and nesting habitat resulted in rapid and persistent occupation by bees, suggesting that efforts to restore bee communities can be successful when a source of colonists exists nearby.

4. Based on earlier studies, we predicted that in restored sites there would be an initial rapid increase in bee abundance and species richness, followed by a decline to a stable intermediate level. This prediction was supported as bee abundance and species richness in restored sites increased until 2006/2007 and subsequently declined.

5. In control sites, there were significant declines in abundance and species richness over time, despite a lack of anthropogenic disturbance. Possible contributors are changing weather patterns, especially severe droughts; plant community succession resulting in loss of bare ground for nesting sites; and increasing suburbanisation of the surrounding landscape.

Key words. Abundance, anthropogenic disturbance, bee community restoration, demographic variation, long-term study, species richness.

Correspondence: Thomas M. Onuferko or Miriam H. Richards, 4700 Keele Street, Toronto, ON M3J 1P3, Canada and St. Catharines, ON L2S 3A1, Canada. E-mail: thomas.onuferko@gmail.com or mrichards@brocku.ca

Introduction

Understanding how organisms, including ecologically and economically important groups like bees, make use of human-provided habitats can be important for conservation planning and the successful restoration of ecosystem services (Williams *et al.*, 2010; Winfree, 2010). Habitat restoration allows new communities to be re-established in areas from which they were previously displaced. Studies of the initial phases of organismal responses to habitat restoration are important because they can demonstrate how and why species return to areas from which they were previously extirpated, but it is also important to observe the long-term results of recolonisation. Unlike short-term studies, long-term studies may reveal which types of species (i.e. taxonomic or functional groups) are most likely to establish themselves successfully in the long run, and whether restored areas end up with species assemblages similar to or differing from source populations and other communities in surrounding areas (Ruiz-Jaen & Aide, 2005).

Short-term demographic responses of bees

The sensitivity of bees to disturbance usually manifests as declines in abundance in response to habitat loss and land use change (Winfree *et al*., 2009; De Palma *et al*., 2016). While sensitive to perturbations, bees can also respond quickly and positively to habitat creation and restoration. The major resources required for bees to establish in particular areas are food and nest sites. Flowers provide nectar and pollen resources used by adult bees to support their own activities and to provision immatures. Some bees are foraging specialists, feeding and gathering larval provisions on particular plant species or groups of species, while others are nesting specialists, with distinct requirements for nesting substrate. For many bees, suitable nesting substrate comprises bare soil patches for excavating underground nests, woody substrates such as twigs and logs for excavating tunnels, or pre-formed holes in wood or stone for cavity nesters. When habitat restoration provides new floral and nesting resources, at least generalist bees should arrive and establish themselves quickly.

Evidence that bees can quickly recolonise sites from which they were previously eradicated comes from studies of areas formerly subjected to severe anthropogenic disturbance, such as sand pits, quarries and road verges. For instance, in north-eastern Kansas, bee abundance and species richness in restored roadside habitat had risen to levels observed in prairie remnants within only 4 years (Hopwood, 2008). Exeler *et al*. (2009) found that bee abundance and species richness increased rapidly in new sand dune and grassland habitats in north-western Germany within a few years of restoration, and by these measures (though not in species composition) the restored sites greatly matched the target sites. In riparian habitats, restored sites contained an abundance of bees and almost as many species as nearby undisturbed remnant sites after just 6 years of recolonisation (Williams, 2011).

Long-term demographic patterns in bee communities

Relatively little is known about long-term demographic variation (i.e. year-to-year variation in abundance and species richness) in bee communities in either restored or preserved sites because of the difficulties of sustained replicated sampling and large amounts of specimen processing associated with long-term monitoring. The few continuous, long-term studies available suggest that high variability in abundance may be a hallmark of bee population dynamics. For instance, a 10-year study of bumblebees in French hay fields demonstrated so much variation in bumblebee abundance that species that were dominant 1 year were close to disappearing in subsequent years (Iserbyt & Rasmont, 2012). In Panama, nocturnal bees (especially *Megalopta* Smith sweat bees) and euglossine bees also exhibited considerable variation in abundance from year to year (Wolda & Roubik, 1986; Roubik, 2001). In a U.S. desert bee community, regionally common species exhibited the highest variation in abundance among years, but were also the most temporally persistent bees (i.e. did not experience turnover during the annual and decadal time scales studied) (Cane *et al*., 2005).

Despite high variance in abundance, bee populations and communities may also be highly persistent over relatively long periods of time (i.e. longer than 10 years). Restoration of British heathlands resulted in plant and pollinator communities as robust as those found in ancient heathlands when examined 11 years after initial restoration management (Forup *et al*., 2008). In before-and-after studies, however, it is a challenge to distinguish long-term patterns (declines, increases or overall stability in bee numbers) from year-to-year variability. If there is significant demographic variation from year to year, then long-term studies based on two temporally separate sampling periods could interpret short-term variance in bee communities as long-term change (Richardson & Richards, 2008). Yet most studies of long-term demographic patterns are based either on comparisons between two time sampling periods separated by a long period of time when no sampling was done (e.g. Marlin & LaBerge, 2001; Grixti & Packer, 2006; Norden, 2008; Kearns & Oliveras, 2009) or on comparisons among bee assemblages in restored habitats of different ages (Hopwood, 2008; Exeler *et al*., 2009; Williams, 2011; Rutgers-Kelly & Richards, 2013).

One way to study long-term demographic patterns is to continually track focal bee communities over time. Studies of bee communities in restored habitats can help to distinguish between short-term and long-term responses to restoration, as well as revealing the extent of year-to-year variation. Several studies suggest that after an initial peak in bee species diversity within a few years of restoration, species richness subsequently declines gradually with ecological succession (Connell, 1978; Steffan-Dewenter & Tscharntke, 2001; Grixti & Packer, 2006; Heneberg *et al*., 2013; Rutgers-Kelly & Richards, 2013). How often bee communities achieve stability and how long 'stability' actually lasts is not clear, but continuous long-term studies provide the information necessary to understand the degree of annual variation in bee community demography so that declines, increases or long-term stability can be better understood.

The landscape of southern Ontario, Canada, including the Niagara Peninsula, has been extensively modified by urbanisation, suburbanisation and intensive agriculture practices over the past 200 years (Reid & Symmes, 1997). Prior to European settlement, much of this region was forest, but there was also tallgrass prairie that would have provided important bee habitat. The grassland that originally covered roughly 1000 km^2 in southern Ontario, however, has been reduced to <3% of its original cover, and few remnants of the original forests or grasslands remain (Reid & Symmes, 1997). Fortunately, there have been efforts to recreate small patches of naturalised habitat that would provide protected areas for plant and animal communities, including pollinators, as well as natural areas for human recreational activity. In particular, the Niagara Regional Government has closed several large landfill sites and redeveloped them as naturalisation sites.

The current study is part of a long-term project focusing on patterns of bee abundance and diversity in meadow habitats of varying ages in southern Ontario (León Cordero, 2011; Richards *et al.*, 2011; Rutgers-Kelly & Richards, 2013). The Glenridge Quarry Naturalization Site (GQNS) is now a park, but for decades was a farm, then a quarry and finally a major landfill site; neither flowers nor nesting sites were available while it was a quarry and landfill, so bees were effectively eradicated until the landfill was closed and construction ceased in 2003 (Richards *et al.*, 2011). When the site was revegetated in 2003, bees immediately began to return to the GQNS (Rutgers-Kelly & Richards, 2013). Initial studies in 2003 compared bee abundance and diversity between newly restored sites, 2- to 3-year-old sites and control sites nearby that had experienced low disturbances for at least 40 years (Richards *et al.*, 2011; Rutgers-Kelly & Richards, 2013). These comparisons demonstrated that by the end of the first year of restoration, bee abundance and diversity were highest in 2- to 3-year-old peripheral sites, lowest in the newest sites and intermediate in the 40-year-old control sites. This pattern suggested that when bees encounter newly available habitat, their abundance and diversity initially rise rapidly for 2–3 years, but then decline to intermediate, possibly steady-state levels (Richards *et al.*, 2011; Rutgers-Kelly & Richards, 2013). Here, we present a 10-year longitudinal study designed to investigate whether the time course of recovery followed the pattern inferred in 2003 from comparisons among sites of different ages. By comparing bee abundance and diversity in sites restored in 2003 to control sites, over 10 years of sampling from 2003 to 2013, we addressed the following objectives and predictions based on the initial studies (Richards *et al.*, 2011; Rutgers-Kelly & Richards, 2013):

1. Our first objective is to investigate the time course of bee community recovery in newly restored sites by examining changes in bee abundance and diversity in restored sites, in comparison with control sites. We predicted that bee abundance and diversity in restored sites would increase to the level of control sites within several years (i.e. by 2005 or 2006).
2. Our second objective is to examine long-term patterns in bee abundance and diversity over the 10 years of sampling (from 2003 to 2013). We predicted that control sites would have stable bee abundance and diversity, having remained relatively undisturbed for decades. In line with the observations of the 2003 census study and the Intermediate Disturbance Hypothesis, we predicted initial rapid recovery of bee populations in restored sites, followed by a slow decline in species richness to intermediate levels.
3. In the course of the study, we uncovered persistent differences between spring and summer bee assemblages. We therefore compared spring and summer bee abundance and diversity across years and restoration levels.

Materials and methods

Study sites

This study compares control sites at Brock University to restored sites at the adjacent GQNS in southern St. Catharines, Ontario, Canada (43.122°N, 79.237°W). The Brock University campus and the GQNS are surrounded on the north and west by the forested cliffs of the Niagara Escarpment, and on the west, south and east by woodland, artificial lakes, roadways, highways and parking lots (Rutgers-Kelly & Richards, 2013). Beyond these are suburbs, with some farmland to the south. There were three control and two restored sampling sites; multiple sites within each 'restoration level' (i.e. site type) were used to improve representation of bee species with short foraging distances and localised nesting habitats.

The three control sites represent areas of relatively low disturbance where bees would have been able to nest and forage uninterrupted for decades (at least since major construction of the university in the 1960s), and which undoubtedly were a major source of bees from which many emigrated to the newly available habitat represented by the restoration sites (Rutgers-Kelly & Richards, 2013). The control sites comprised two adjacent fields at the southern end of the Brock University campus (BrNW and BrS), separated by a row of trees and shrubs and with a distance of about 100–150 m between linear transects, and a third field (406) between the GQNS and Highway 406.

The two restored sites were located on the northern [south of a large artificial pond (Pon)] and eastern [adjacent to St. David's Road (StD)] slopes of a small hill in the GQNS, in areas that were rock quarry until about 1971 and subsequently the centre of the landfill until it was covered in 2001. After the landfill was capped with clay and soil, it was replanted in the spring of 2003. Thus, the restored sites, in which transects were separated by a distance between 150 and 200 m, provided neither floral nor nesting resources prior to that year (Rutgers-Kelly & Richards, 2013).

Pan-trapping methods

Each year from 2003 to 2013 (except 2007, when no bees were collected), bees were sampled using pan traps. Pan traps were used in order to standardise collecting effort as much as possible among years and collectors. At least one control and one restored site were sampled each year (Table S1). Pan traps were 170 g Solo brand PS6-0099 plastic bowls (Solo Cup Company, Lake Forest, IL, USA) in three colours: white (unpainted), fluorescent yellow (Krylon paint brand #3104) and blue (#3109). Pans were filled with water and a small amount of blue liquid detergent ('Dawn'). Pan traps were set out by 09.00 hours and recovered by 16.00 hours of the same collecting day. Sampling was done on clear days when bees were expected to be foraging. If pan traps were set out and unexpectedly it started to rain, the samples were replaced on a subsequent day without rain. Pan trapping was carried out at least biweekly, from April to September, weather conditions permitting.

Over the course of the 10-year study, the shapes of pan trapping transects were changed. All transect shapes used 30 pans, 10 blue, 10 white and 10 yellow, laid out in alternating colours. From 2003 to 2008, pans were set out in space-filling patterns that varied with the shapes of the sampling sites (Richards *et al.*, 2011; Rutgers-Kelly & Richards, 2013), whereas from 2009 to 2013, pans were set out in a single line. Space-filling transect shapes in each site were as follows. From 2003 to 2006, sites 406, BrNW and BrS were sampled with two crossed lines (each 140 m) forming an X-shape, with 10 m between pans. From 2003 to 2005, sites Pon and StD were sampled in an H-shape, which was formed with one 140-m line crossed by two parallel lines of unequal length (70 and 80 m). In 2006, these sites were sampled with two 140-m parallel lines separated by 20 m, with pans 10 m apart. In 2008, sites were sampled using an X-shaped transect pattern consisting of two 50-m lines, with 3.3 m between pans. From 2009 onwards, all traps were set out along a straight line with 3.3 m between pans to match the Canadian Pollinator Initiative sampling protocol (www.uoguelph.ca/canpolin/Sampling/protocols.html). We categorised transect types as space filling (2003–2008) or linear (2009–2013). Sites were usually sampled biweekly, but in some years were sampled weekly. Week numbers were assigned using the WEEKNUM function in Excel, with week 0 being in late April. Over the 10 years of sampling from 2003 to 2013, there were 540 pan trap collections, each comprising all bees trapped at a site in a single day (Table S1).

Bee identifications

The Niagara region of southern Ontario is home to at least 149 bee species in five families (Andrenidae, Apidae, Colletidae, Halictidae and Megachilidae; Onuferko *et al.*, 2015), of which 110 morphospecies, representing at least 112 species, have been caught in pan traps at our five study sites. A total of 23 282 specimens were collected in pan traps over the 10 years. Onuferko *et al.* (2015) provides a detailed description of the curation methods used in the present study. Identification methods are indicated in Table S2. In summary, identifications were made with the help of taxonomic experts (Sam Droege, Jason Gibbs and Cory Sheffield) and using the following keys: Mitchell (1960, 1962), McGinley (1986), Packer *et al.* (2007) Rehan and Richards (2008), Rightmyer (2008), Gibbs (2010, 2011), Colla *et al.* (2011), Rehan and Sheffield (2011), Sheffield *et al.* (2011), Gibbs *et al.* (2013) and Discover Life (Ascher & Pickering, 2015). We use the term 'species' inclusively to refer to both species and morphospecies. The following three were treated as individual morphospecies in our analyses of species richness: bidentate species of *Nomada* Scopoli (Apidae), *Hylaeus affinis* (Smith) (Colletidae) and *H. modestus* Say, and *Ceratina dupla* Say (Apidae) and *C. mikmaqi* Rehan and Sheffield, which were difficult or impossible to tell apart due to problematic taxonomy. For the most part, Andrenidae were not identified to species after 2003, so andrenids were included in analyses of abundance, but not species richness. Damaged specimens that could not be identified to species or morphospecies were excluded from analyses (Table S2).

Data analysis

Variable selection. Bee abundance was measured as the total number of bees per collection, and diversity was measured as the number of species per collection. This normalised both abundance and species richness to collection effort, which differed among sites, seasons and years; pooling collections by sites, seasons and years would be a biased measure. Previous studies at our sites and in other regions suggested that temperate bee communities comprised distinct assemblages based on bees that forage and breed in spring versus summer (Oertli *et al.*, 2005; Minckley, 2008; Richards *et al.*, 2011). At our sites there was a hiatus in bee foraging activity in late June to early July each year, so we designated spring as weeks 0–10, and summer as week 11 onward.

Multivariate analyses. To investigate how species abundance patterns of Apidae, Colletidae, Halictidae and Megachilidae changed over time and whether the patterns were affected by bee family, season and restoration level, we used permutational multivariate analysis of variance (PERMANOVA), as implemented in the adonis function in the R package vegan. The response variable for these analyses was species-level abundance per collection. Andrenidae were not included in multivariate analyses, as andrenids were not identified to species after 2003. We tested the effects of year, season, restoration level, family and all possible interactions on the number of bees of each species in each collection, including zeros for bees not collected. Year, season, restoration level, family and transect type were treated as fixed factors, and site was treated as a random factor (note that the *adonis* function allows only one random variable). To more closely examine compositional differences among four bee families, we also subsetted the data by family and carried out the same analyses.

Generalised additive mixed modelling approaches. We used generalised additive mixed models (GAMM) to examine sources of variation in bee abundance and species richness, because this allowed us to relax the assumption that changes in bee abundance over time were linear. This was particularly relevant for restored sites, for which our previous studies (Richards *et al.*, 2011; Rutgers-Kelly & Richards, 2013) predicted several years of increasing abundance, followed by a decline or stabilisation in numbers. For GAMM analyses of abundance, the response variable was number of bees per family per collection, so Andrenidae were included. Abundance was log transformed to improve model fit (see Ives, 2015) and modelled through a Gaussian distribution with an identity link. Species richness was modelled through a quasi-Poisson distribution with a natural logarithm link. We treated season, family and restoration level as fixed effects with all possible interactions, and transect type, site and week as random effects. Transect type was included because space-filling transect shapes tend to trap more specimens per pan (Droege *et al.*, 2010). Because yearly and weekly patterns of abundance and richness are continuous and correlated (Richards *et al.*, 2011; Rutgers-Kelly & Richards, 2013), we modelled both year and week by smooth functions. Separate tensor product smooths were calculated for each combination of year and restoration level, and for week [*gamm*() in mgcv, version 1.8-4, Wood, 2011, which calls the *lme*() function, and that the models were based on reduced maximum likelihood.]. To reduce the influence of overfitting on the GAMM, year smooths were restricted to a basis spline dimension of $k = 4$ in abundance models and $k = 3$ in species richness models. We implemented an autoregressive moving average to account for autocorrelations among weeks.

Rarefaction curves. We used EstimateS version 9.0.0 (Colwell, 2013) to construct individual-based rarefaction curves. Rarefaction curves were estimated based on 100 randomisations (the recommended default number) without replacement to ensure that the maximum number of species estimated matched the actual maximum number of species observed.

Results

During the 10 years of pan trapping from 2003 to 2013, we collected 23 282 specimens of at least 112 morphospecies in the families Andrenidae, Apidae, Colletidae, Halictidae and Megachilidae (Table S2). Within collections and across the whole data set, the total number of bees per species varied over three orders of magnitude. The two most abundant species [the metallic green sweat bee *Augochlorella aurata* (Smith) (Halictidae) and the snail shell-nesting mason bee *Osmia conjuncta* Cresson (Megachilidae)] comprised 32% and 16% of all sampled specimens, respectively, and the 10 most common morphospecies (representing 12 species) represented 87% of sampled individuals.

Multivariate analyses

Multivariate analyses of abundance from 2003 to 2013 demonstrate considerable variability in species abundance (Table 1). The strongest contributor to variation in bee abundance was the main effect of family ($r^2 = 52\%$) (differences among families are illustrated in Fig. S1). Year, season and restoration level also contributed significantly to overall variation, whereas transect type had no significant effect. All two- and three-way interaction effects involving family were significant, most notably the interaction between season and family ($r^2 = 5\%$).

Separate PERMANOVA analyses on Apidae, Colletidae, Halictidae and Megachilidae (Table 2) provide further insight into variation among bee families. Only Halictidae exhibited a strong main effect of year, although the effect for Colletidae was close to significant ($P = 0.053$, Table 2). All four families exhibited significant seasonal effects (i.e. differences between spring and summer). Three families, Apidae, Halictidae and Megachilidae, showed significant interactions between year and season, suggesting that changes over the years differed in the spring and summer components of these assemblages. Apidae, Halictidae and Megachilidae also exhibited significant differences between restoration levels (control vs. restored). In Megachilidae, there was also a significant interaction between season and restoration level ($P = 0.035$), suggesting that the abundance of bees in control versus restored sites depended on the season.

Generalised additive mixed modelling of abundance

The significant variation in bee abundance revealed by the above multivariate approach reflects abundance variation both within and among species, as analysed by models based on assumptions of, most critically, linear patterns over years. The GAMM modelling of bee family abundance per collection (Table 3) reproduced several patterns found by multivariate analyses of species-level abundance. Using the GAMM, the terms that explained the most variation in bee abundance were family and the interaction between season and family (Table 3), as was also true of the PERMANOVA (Table 1). The three-way interaction including season, restoration level and family was also significant. Unlike the PERMANOVA, the GAMM suggested that there was no main effect of season, but in the latter case this variation is instead incorporated into the week smooth, which accounted for significant

Table 1. Permutational multivariate analysis of variance summary results for variation in the abundance of individual species sampled in control and restored sites from 2003 to 2013.

Partial effects	d.f.	F	r^2	P
Year	1	3.644	0.004	0.001***
Season	1	14.472	0.018	0.000***
Restoration level	1	2.458	0.003	0.000***
Family	3	145.008	0.527	0.000***
Transect type	1	1.301	0.002	0.218
Year : Season	1	2.358	0.003	0.019*
Year : Restoration level	1	0.942	0.001	0.441
Year : Family	3	3.558	0.013	0.000***
Season : Restoration level	1	2.121	0.003	0.031*
Season : Family	3	14.237	0.052	0.000***
Restoration level : Family	3	2.516	0.009	0.000***
Year : Season : Restoration level	1	1.125	0.001	0.322
Year : Season : Family	3	2.354	0.009	0.001***
Year : Restoration level : Family	3	0.963	0.004	0.512
Season : Restoration level : Family	3	2.085	0.008	0.003**
Year : Season : Restoration level : Family	3	1.123	0.004	0.305
Residuals	282	0.341		
Total	314			

Interaction effects are indicated by a colon.
$***P < 0.001$, $**P < 0.01$, $*P < 0.05$.

variation in abundance, including an overall decline in abundance from spring to summer (Fig. 1).

The GAMM analysis also did not produce a main effect of restoration level, but produced significant evidence that year-to-year abundance patterns differed among seasons and restoration levels (Fig. 2; Table 3). Spring bees declined over time in control sites, whereas in restored sites, they increased in abundance for 3–4 years, before beginning to decline around 2007. Summer bees also declined significantly over time in control sites, whereas in restored sites, their numbers increased slightly for about 3 years, before declining or stabilising at slightly lower levels, starting around 2006. Thus, in restored sites, both spring and summer bee abundance rose for several years in response to restoration, then seems to have declined to more or less stable numbers from around 2009 to 2013.

Generalised additive mixed modelling of species richness

Neither season nor restoration level contributed significantly to variation in species richness, either as main or interaction effects. The lack of an overall difference in species richness was confirmed by rarefaction curves (Fig. 3), demonstrating no perceptible differences between control and restored sites when numbers were pooled for the entire 2003–2013 study period.

The GAMM smoothing terms indicated significant temporal variation in species richness. Within years, species

Table 2. Permutational multivariate analysis of variance summary results for variation in the abundance of individual species (analysed separately for each of four famlieis for which all specimens were identified to species) sampled in control and restored sites from 2003 to 2013.

Partial effects	d.f.	F	r^2	P	Partial effects	d.f.	F	r^2	P
Apidae					Colletidae				
Year	1	1.344	0.014	0.227	Year	1	2.693	0.033	0.053
Season	1	11.030	0.116	0.000***	Season	1	6.259	0.076	0.002**
Restoration level	1	2.778	0.029	0.000***	Restoration level	1	0.818	0.010	0.916
Transect type	1	1.565	0.016	0.164	Transect type	1	1.469	0.018	0.212
Year : Season	1	3.312	0.035	0.013*	Year : Season	1	1.029	0.012	0.370
Year : Restoration level	1	0.481	0.005	0.826	Year : Restoration level	1	0.230	0.003	0.890
Season : Restoration level	1	2.354	0.025	0.060	Season : Restoration level	1	1.513	0.018	0.201
Year : Season : Restoration level	1	1.114	0.012	0.370	Year : Season : Restoration level	1	0.546	0.007	0.619
Residuals	71	0.748			Residuals	68	0.824		
Total	79				Total	76			
Halictidae					Megachilidae				
Year	1	9.603	0.103	0.000***	Year	1	1.762	0.015	0.100
Season	1	2.407	0.026	0.045*	Season	1	32.877	0.280	0.000***
Restoration level	1	2.100	0.023	0.000***	Restoration level	1	4.183	0.036	0.000***
Transect type	1	0.967	0.010	0.387	Transect type	1	1.286	0.011	0.248
Year : Season	1	2.935	0.031	0.022*	Year : Season	1	2.432	0.021	0.039*
Year : Restoration level	1	1.638	0.018	0.158	Year : Restoration level	1	1.565	0.013	0.154
Season : Restoration level	1	1.868	0.020	0.097	Season : Restoration level	1	2.557	0.022	0.035*
Year : Season : Restoration level	1	0.726	0.008	0.554	Year : Season : Restoration level	1	1.952	0.017	0.088
Residuals	71	0.761			Residuals	69	0.587		
Total	79				Total	77			

Interaction effects are indicated by a colon.
$***P < 0.001$, $**P < 0.01$, $*P < 0.05$.

Table 3. Generalised additive mixed model analysing variation in bee abundance (bees per family per collection, log-transformed).

Parametric terms	d.f.	*F*	*P*
Season	1	0.000	0.989
Restoration level	1	0.299	0.584
Family	4	127.124	0.000
Season : Restoration level	1	2.433	0.119
Season : Family	4	21.418	0.000
Restoration level : Family	4	3.086	0.015
Season : Restoration level : Family	4	3.563	0.007

Smoothing terms	E.d.f.	Ref. d.f.	*F*	*P*
Year:				
Spring : Control	1.384	3.000	3.293	0.004
Spring : Restored	2.819	3.000	12.280	0.000
Summer : Control	1.809	3.000	8.231	0.000
Summer : Restored	2.654	3.000	8.474	0.000
Week	2.951	2.951	23.315	0.000

Interaction effects are indicated by a colon. Overall r^2(adj) = 35.4% (n = 1829 observations).
E.d.f., estimated degrees of freedom for the model terms; Ref. d.f., degrees of freedom for reference distributions.

richness declined towards the end of the summer (Fig. 1b). Among spring bees, there was no change in species richness over time in the control sites (Fig. 4; Table 4), while in the restored sites, there was a small increase following restoration (although the curve is very close to flat). By contrast, among summer bees, there were significant declines in species richness in both the control and restored sites.

Discussion

To our knowledge, the present study represents the longest survey of annual variation in the abundance and species richness of a bee community, and the only one to track restoration continually from the very beginning of the process of restoration and recolonisation of habitat. Hopwood (2008) and Exeler *et al.* (2009) studied restoration that occurred through secondary succession, in which one type of habitat was converted into another, so bees were probably never fully eradicated. Exeler *et al.* (2009) sampled bees immediately after restoration, but not in the subsequent year (longitudinal comparisons were made to samples taken 2 and 3 years later). In our study, we found two different patterns. First, as predicted, there was convincing evidence for rapid re-establishment of the bee community by about 2006 or 2007, 3 to 4 years after full restoration of the former landfill sites. Second, not as predicted, there was strong evidence for a long-term decline in bee abundance from 2003 to 2013 in the relatively undisturbed control sites that were used as a reference for understanding year-to-year variation in restored sites. Third, there was good evidence that spring and summer bee assemblages are demographically distinct, differing both in responses to restoration and in long-term demographic patterns.

Rapid recovery of bee assemblages in restored sites

There are two basic approaches to investigating the time course of ecological recovery in restored sites. The cross-sectional approach, which has been used by many studies, simultaneously censuses bee assemblages in habitats of different ages (reviewed in De Palma *et al.*, 2016). This approach was used in our initial studies in 2003 (Rutgers-Kelly & Richards, 2013). Comparing then newly restored (0–5 months after restoration), intermediate (2- to 3-year-old) and decades-old (control) sites provided strong, but indirect evidence that recovery of bee communities in newly restored habitat could proceed very quickly, within as little as 3 years. In the current study, the longitudinal approach, in which we tracked restored sites from the initial stages of recovery, suggested that

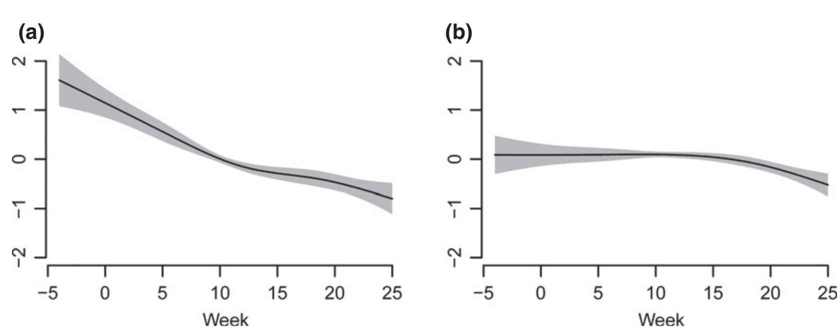

Fig. 1. Generalised additive mixed model splines fitting weekly changes in bee family abundance (a) and bee species richness (b) per collection from the start of the first sampling week (−4 in March) to week 25 (in October). The shaded areas around the curves represent standard errors. The y-axis represents fitted values of log(Abundance) (a) and species richness (b) on the scale of the identity link function based on a Gaussian distribution.

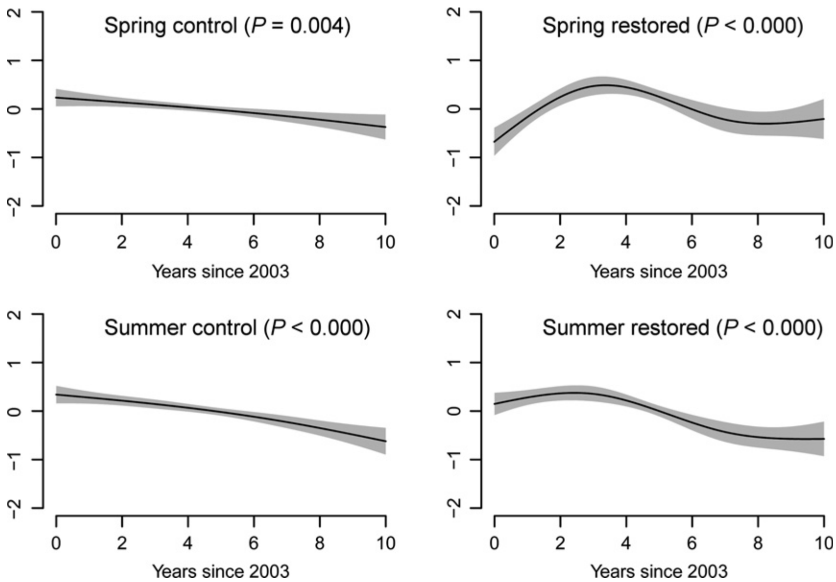

Fig. 2. Generalised additive mixed model splines fitting yearly changes in bee family abundance per collection from the start of the study (0 = 2003) to year 10 (2013). The shaded areas around the curves represent standard errors. In all figures, the *y*-axis represents fitted values of log(Abundance) on the scale of the identity link function based on a Gaussian distribution. Probabilities indicate smooths that are significantly different from a flat line (see Table 3).

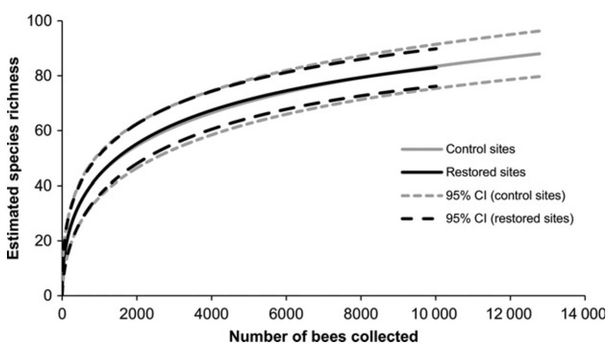

Fig. 3. Rarefied bee species richness (*S*) (standardised for variation in the number of bees collected) in control and restored sites. Andrenidae were not included in this analysis. EstimateS version 9.0.0 (Colwell, 2013) was used for individual-based rarefaction, in which the data set was resampled 100 times without replacement.

recovery was about as fast as we predicted, as bee abundance and species richness increased for several years until 2006. The rarefaction analyses of all 10 years of data demonstrated that the control and restored sites were virtually indistinguishable by the end of this study period. In fact, from about 2006 or 2007, bee abundance and species richness in restored sites either stabilised at intermediate levels, as predicted by Rutgers-Kelly and Richards (2013) or was beginning to decline in concert with the declines noted in control sites. This suggests that when viable source populations exist nearby (such as those in the contiguous control sites), bees rapidly immigrate into newly available habitat. An important question is whether the 3- to 4-year time course of recovery in our study was specific to our particular research sites and habitats, or whether bee communities generally reassemble quickly when reclaimed habitat becomes available in areas where there are viable source populations of potential emigrants. Studies in temperate regions of restoration through conversion of one type of habitat into another found that secondary succession resulted in comparable levels of species richness within 2 years (Exeler *et al.*, 2009) and 4 years (Hopwood, 2008).

Our earlier study suggested that after the initial phase of recolonisation, bee abundance and diversity in restored sites would eventually decline to levels seen in control sites, consistent with the predictions of the Intermediate Disturbance Hypothesis (Rutgers-Kelly & Richards, 2013). The smoothed year effects for both abundance and (in spring bees) species richness certainly suggest restoration resulted in increases in both variables. Subsequently, it also seems likely that both abundance and species richness stabilised at intermediate levels in restored sites. This trend coincides with the predictions of the Intermediate Disturbance Hypothesis (Connell, 1978), but it is not clear if the shapes of the smooths from 2008 onward represent stability or slow changes. This may become clearer as the study is extended. Given the dearth of long-term studies on bee communities or populations, it is difficult to know how variable bee abundance really is from 1 year to the next, but a better understanding of year-to-year variation may alter the conclusions from before-and-after comparisons of historically sampled sites (e.g. Marlin & LaBerge, 2001; Grixti & Packer, 2006; Norden, 2008; Kearns & Oliveras, 2009).

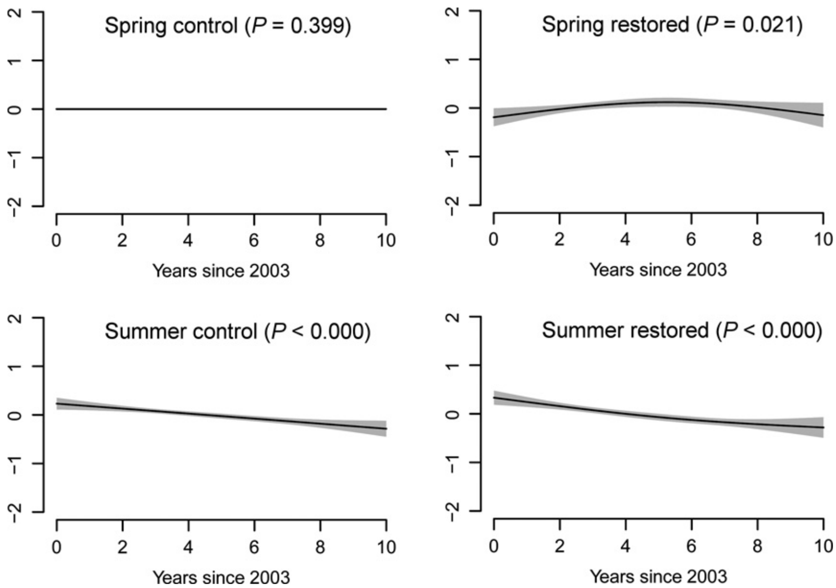

Fig. 4. Generalised additive mixed model splines fitting yearly changes in bee species richness from the start of the study (0 = 2003) to year 10 (2013). The shaded areas around the curves represent standard errors. In all figures, the *y*-axis represents fitted values of species richness per collection on the scale of the logit link function based on a quasi-Poisson distribution. Probabilities indicate smooths that are significantly different from a flat line (see Table 4).

Table 4. Generalised additive mixed model analysing variation in bee species richness.

Parametric terms		d.f.	*F*	*P*
Season		1	0.158	0.691
Restoration level		1	0.255	0.614
Season : Restoration level		1	2.834	0.093
Smoothing terms	**E.d.f.**	**Ref. d.f.**	***F***	***P***
Year:				
Spring : Control	0.000	2.000	0.000	0.399
Spring : Restored	1.707	2.000	12.280	0.021
Summer : Control	1.376	2.000	8.231	0.000
Summer : Restored	1.509	2.000	8.474	0.000
Week	2.625	2.625	23.315	0.000

Interaction effects are indicated by a colon. Overall r^2(adj) = 12.2% (n = 533 observations).
E.d.f., estimated degrees of freedom for the model terms, Ref. d.f., degrees of freedom for reference distributions.

Long-term decline in control sites

The long-term declines in bee abundance and species richness in control sites from 2003 to 2013 were surprising because all sites were protected from disturbances such as mowing and herbicide or insecticide spraying. The declines in abundance in control sites were evident in both spring and summer bees (Fig. 2). Only summer bees also exhibited declines in species richness in the control sites (Fig. 4). A possible cause of the long-term declines in bee abundance at our sites was drought, which would have a greater impact on bees that forage in summer. Between 2003 and 2013, there were two significant summer droughts, one in 2007 and one in 2012. Both droughts had obvious negative effects on plant growth in both natural and agricultural ecosystems, and undoubtedly contributed to lower foraging success by bees. For instance, in 2012, small carpenter bees (*Ceratina calcarata* Robertson) foraged less efficiently and produced fewer broods than in 2013, when rainfall was normal (Lewis & Richards, 2017). Since lack of food in 1 year can result in both immediate and delayed demographic effects on bee populations (Crone, 2013), we surmise that the 2007 drought might have contributed to the low abundance and diversity of bees in 2008, while the drought of 2012 likely led to low abundance in that year and in 2013. Iserbyt and Rasmont (2012) found that the abundance of several bumble bee species declined after ‘hot and dry events’ (droughts), and these effects were particularly marked in the following year. Unusually cool or rainy conditions in late spring may also lead to abundance declines, because the broods of ground-nesting bees are susceptible to rot when rains occur frequently (Packer, 1988; Richards & Packer, 1995). Thus, cool, rainy conditions in late spring and early summer in 2004 likely explain lower bee abundance that year compared to 2003.

While changing weather conditions may be responsible for some of the declines in bee abundance in our research sites, another possibility is changes in the surrounding landscape. When we began the study in 2003, there were highways but little urban development to the south of our study areas (Richards *et al.*, 2011), but over the course of the study, there was considerable development on or beside the control sites at Brock University, as buildings and parking lots were constructed. Although our particular sites were not heavily disturbed, the loss of habitat around the sites has likely contributed to declining populations in the control sites. Evidence from the latter half of our study, namely from 2008 to 2013, suggests that bee populations in restored sites either stabilised or are declining as well. When we began our study, there were not only roads and highways around the restored sites, but also large open areas containing meadows and agricultural areas. These have largely disappeared as new suburbs have been built, so larger landscape effects may be influencing the demography of bees in the restored sites, despite the fact that these sites are protected.

Analyses based on GAMM smoothing suggest that spring bee abundance declined over time in control sites, but species richness did not. In restored sites, species richness also did not change very much over time (slight inverted U shape, Fig. 4). We suspect that declines in abundance largely reflect declines in the numbers of the two most common species, *A. aurata* and *O. conjuncta*, which would not affect species richness. It could be argued that measuring species richness as the number of species per sample, underestimated overall bee diversity, because this measure does not reflect species turnover among samples, sites or years. That there was some level of species turnover is suggested by the fact that a few species caught in 2003 (*Ceratina strenua* Smith, *Nomada bethunei* Cockerell, *N. cressonii* Robertson, *Anthidiellum notatum* (Latreille), and possibly one or more *Andrena* Fabricius species) were never caught again, while new species appeared in later years (Sheffield *et al.*, 2010; Onuferko, 2014). Other studies have also provided evidence for successional change in bee communities in sites shielded from anthropogenic disturbance (Steffan-Dewenter & Tscharntke, 2001; Grixti & Packer, 2006; Heneberg *et al.*, 2013).

Spring and summer bees: two distinct assemblages

Both the PERMANOVA and GAMM approaches demonstrated considerable temporal variation in bee abundance within years. In the multivariate analyses, this was expressed as significant differences between spring and summer bees. In the GAMM analyses, this was expressed not only as week to week variability, but also as distinct year-to-year patterns in abundance and species richness of spring and summer bees. The separation between spring and summer bee communities partly reflects the timing of bee emergence from hibernal diapause, and also reflects the timing of brood production. At our sites, there is a distinct lull in bee flight activity shortly after the summer solstice (Richards *et al.*, 2011), which marks the end of brood provisioning by spring bees and the onset of activity by summer bees, including the numerous workers of eusocial halictids. The distinction between temperate spring and summer bee communities is likely a general pattern, having also been noted in mesic temperate bee communities in Carlinville, IL, USA (Minckley, 2008) and in the Swiss Alps (Oertli *et al.*, 2005).

Differences among bee families

Multivariate analyses demonstrated very different species responses among the four families in which specimens were identified to species. The effect of season was strong in all four families, as expected, given the different phenologies and thus abundance patterns of different types of bees. The only family showing a highly significant main effect of year was Halictidae, the most common bees in our study sites, and thus a strong influence on the demography of the bee community considered as a whole. Most halictids at our sites are ground-nesting, eusocial species and are part of both seasonal assemblages, with queens appearing in spring and workers appearing in summer. The most abundant bee at our sites, *A. aurata*, is a eusocial halictid that was much more abundant at the beginning of our study than it was by the end. Eusocial sweat bees may be particularly susceptible to extreme weather events, such as flooding in late spring or drought in summer (discussed above). It is most likely that their decline over time was due in large part to a loss of bare soil patches for nesting sites, especially in control sites, where shrubs and thick grasses grew rapidly after mowing ceased (M. H. Richards, pers. obs.).

Three of the four bee families exhibited significant main effects of restoration level; by the end of the study, apid, halictid and megachilid abundance appeared to be declining in control sites, but stable or increasing in restored sites (M. H. Richards, unpubl. data). As noted above, there were several possible factors contributing to halictid declines, including vegetational succession. This may also have contributed to changes in apid and megachilid abundance. For instance, the most abundant megachilid at our sites, *O. conjuncta*, nests in empty shells of introduced snails [*Cepaea* Held sp. (Helicidae)], which in recent years seem to have become more abundant in restored sites. Possibly snail populations, or the ability of bees to find the empty shells, declined in the control sites as the vegetation became thicker and deeper. Apidae and Megachilidae are long-tongued bees, and may be better adapted than many short-tongued bees at exploiting nectar from flowers with deeper corollas (Rodríguez-Gironés & Santamaría, 2006), such as sweet white clover [*Melilotus* Mill. sp. (Fabaceae)], which at least in the early years of our study was much more abundant in restored sites and attracted mainly long-tongued bees, in particular, species of *Ceratina* Latreille.

Conclusions

The major objectives of this study were to determine how long it takes for a bee community to be fully restored when habitat is recreated and, at the same time, to examine long-term patterns in abundance and diversity. In a world filled with news of declining wild bee populations due to anthropogenic disturbance, our results suggest that 'If you restore it, they will come': restored foraging and nesting sites were re-occupied by bees as soon as they became available, then bee numbers continued to grow for 3–4 years. Eventually, bee abundances in restored areas declined to intermediate levels that represent either longer term equilibrium or eventual progression to the same state as control populations, which were declining by the end of the study. Our results suggest that in restoring bee communities, it would be appropriate to focus on measures that increase and then maintain bee abundance. Thus, monitoring aggregate bee abundance may be a useful shortcut to monitoring the health of bee populations and communities in habitat restoration projects for which the taxonomic expertise required for identification to species is not available. We intend to test the generality of this observation by continuing our long-term studies in these and additional restored landfill sites.

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Supporting Information

Additional Supporting Information may be found in the online version of this article under the DOI reference: doi: 10.1111/icad.12261:

Figure S1. Partial effect of family on bee abundance as calculated by the generalised additive mixed models (GAMM) analyses of Table 3.

Table S1. The numbers of weekly collections at each site from 2003 to 2013.

Table S2. Total numbers of specimens of each species and morphospecies collected in pan traps in Control and Restored sites, 2003–2013 (not including 2007).

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